SQL

SQL: ESSENTIAL TRAINING

DATA ← comes from "DATUM" ≙ piece of information

DATABASE: Collection of DATA; usually organized in tables

TABLE: made up of rows & columns
(↔) (↕)

ROW: each represents a SINGLE piece of data called a Record

COLUMN: each column represents a specific attribute of that data

SQL Special language
- **S**tructured
- **Q**uery
- **L**anguage

RDBS: **R**elationship **D**ata**b**ase **S**ystem

COMMANDS / KEYWORDS

SELECT: Select column name of a table

FROM: specify which table

ORDER BY: specify which column to order: ASC (ascending), DESC (descending)

LIMIT: specify how many rows of data you need

AS: assign name to a column

Basic Query Composition Example

```
SELECT
    ID AS "Customer ID" or [Customer ID],
    FirstName AS Name (when is just one word, no "" or [] needed.
FROM
    Customer
ORDER BY
    Firstname,
    ID DES
LIMIT 50
```

OPERATOR TYPES

ARITHMETIC

Add	+
Subtract	−
Multiply	*
Divide	/
Modulo	%

COMPARISON

Equal to	=
Not equal to	<>
Greater than	>
Less than	<
Greater than or equal to	>=
Less than or equal to	<=

LOGICAL

And
Or
In
Like
Between

WHERE: for filtering data with use of tors
- before ORDER BY

Filtering numeric data

```
SELECT
    InvoiceDate, BillingCity, Total
FROM
    Invoice
WHERE
    total = 1,98
```

Filtering multiple numeric data

```
SELECT
    InvoiceDate, BillingCity, Total
FROM
    Invoice
WHERE
    total IN (1,98, 3,96)  - for exact match
        or
WHERE
    total BETWEEN 1,98 AND 3,96
```

Filtering multiple text data

```
SELECT
    InvoiceDate, BillingCity, Total
FROM
    Invoice
WHERE
    Billingcity IN ('Brussels', 'Orlando', 'Paris')
```

Filtering text data without exact match

```
SELECT
    InvoiceDate, BillingCity, Total
FROM
    Invoice
WHERE
    Billingcity LIKE ('B%')   (NOT LIKE )
```

% : does not care what comes next if placed next to char or what comes before if placed before char.

Filtering dates

```
SELECT
    InvoiceDate, BillingCity, Total
FROM
    Invoice
WHERE
    date(InvoiceDate) = '2010-05-22'
```

DATE() : selects just date w/o time from a DATETIME type

Filtering multiple conditions

```
SELECT
    InvoiceDate, BillingCity, Total
FROM
    Invoice
WHERE
    date(InvoiceDate) > '2010-05-22' (OR) AND total < 3.00
```

Order rules weight!!
PEMDAS - Parenthesis, Exponents, Multiplication/Division, Addition/Subtraction

Adding new column to data using if then logic

CASE : Command to start an IF THEN logic new column
- needs **END** at the end
- can be renamed using as after END
- must be part of **SELECT**

Example

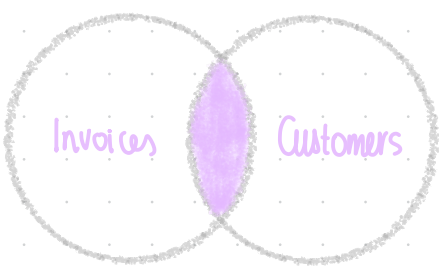
```
SELECT
    InvoiceDate, BillingAdress, BillingCity, total,
CASE
    WHEN total < 2.00 THEN 'Baseline Performer'
    WHEN total < 7.00 THEN 'Target Performer'
    ELSE 'Top Performer'
END
FROM
    Invoice
```

Accessing Data from Multiple Tables

```
SELECT *
FROM
    Invoice AS i
XXX JOIN
    Customer AS c
ON
    i.customerId = c.customerId
```

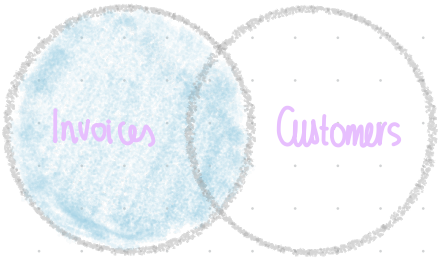
* : c.FirstName
c.Last ame
i.InvoiceId
i.CustomerId
i.InvoiceDate
i.total

INNER JOIN: Only the overlapping section



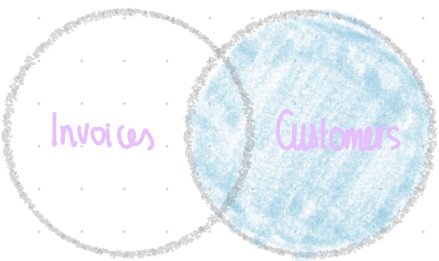

```
SELECT
FROM invoices i
INNER JOIN customers c
ON i.customerId = c.customerId
```

LEFT OUTER JOIN:




```
SELECT
FROM invoices i
LEFT OUTER JOIN customers c
ON i.customerId = c.customerId
```

RIGHT OUTER JOIN:




```
SELECT
FROM invoices i
RIGHT OUTER JOIN customers c
ON i.customerId = c.customerId
```

MORE THAN ONE TABLE JOIN

```
/*
Created by: Andrea Contreras
Created: 22.01.2025
Description: JOINS on more than 2 Tables | What employees are responsible for the 10 highest invoices?
*/

SELECT
    e.FirstName,
    e.LastName,
    e.EmployeeId,
    c.FirstName,
    c.LastName,
    c.SupportRepId,
    i.CustomerId,
    i.total
FROM
    Invoice i
INNER JOIN
    Customer c
ON
    i.CustomerId = c.CustomerId
INNER JOIN
    Employee e
ON
    c.SupportRepId = e.EmployeeId
ORDER BY
    i.total DESC
```

FUNCTIONS / FORMULAS

AGGREGATE

MIN( )
MAX( )
AVG( )
COUNT( )
SUM( )

STRING

UPPER( )
SUBSTR( )
INSTR( )
LENGTH( )
LTRIM( )
REPLACE( )
TRIM( )

DATE

NOW
DATETIME( )
DATE( )
JULIANDAY( )
STRFTIME( )
TIME( )

String Functions

```
SELECT
   FirstName,
   LastName,
   Address,
   FirstName || ''||LastName|| '' || Address || ',' || City || '' || State||''||PostalCode as [Mailing Adress], -- || works like & in Excel
   length(PostalCode) ,
   substr(postalcode,1,5) as [5 Digit Postal Code],
   upper(FirstName) as [ First Name ALL Caps],
   lower(LastName) as [Last Name ALL Lower]
FROM
   Customer
WHERE
   Country = 'USA'
```

irstNam	.astNam	Address	Mailing Adress	Post	5 Digit Postal Code	First Name ALL Caps	Last Name ALL Lower
rank	Harris	1600 ...	Frank Harris 1600 Amphitheatr...	10	94043	FRANK	harris
ack	Smith	1 Microsoft ...	Jack Smith 1 Microsoft ...	10	98052	JACK	smith

Date Functions

```
6   SELECT
7      LastName,
8      FirstName,
9      BirthDate,
10     strftime('%Y-%m-%d',BirthDate) as [Birthday No Timecode],
11     strftime('%Y-%m-%d','now') - strftime('%Y-%m-%d',BirthDate) as Age,
12     date('now')-date(birthdate) as [Age using date() function]
13  FROM
14     Employee
15
```

	LastName	FirstName	BirthDate	Birthday No Timecode	Age	Age using date() function
1	Adams	Andrew	1962-02-18 00:00:00	1962-02-18	63	63
2	Edwards	Nancy	1958-12-08 00:00:00	1958-12-08	67	67

Aggregate Functions & Nesting.

```
SELECT
    sum(total) as [Total Sales],
    round(avg(total),2) as [Average Sales],
    max(total) as [Maximum Sale],
    min(total) as [Minimum Sale],
    count(*) as [Sales Count]
FROM
    Invoice
```

Total Sales	Average Sales	Maximum Sale	Minimum Sale	Sales Count
3329.46	8.06	1000.86	0.99	413

GROUPING

GROUP BY: Often used with aggregate functions

```
Description: Grouping in SQL |What are the average invoice totals by city?
*/
SELECT
    BillingCity, -- non aggregated field
    round(avg(total),2) -- aggregated field
FROM
    Invoice
GROUP BY -- rule of thumb is to group by a non aggregated field
    BillingCity
ORDER By
    BillingCity
```

How do i filter aggregate or non aggregate fields?

Using

- **WHERE** for non aggregated fields
  ALWAYS **BEFORE** Group By
- **HAVING** for aggregated fields
  ALWAYS **AFTER** Group By

Grouping with WHERE & HAVING Clause

```
Description: Grouping in SQL |What are the average invoice totals greater than $ 5.00 in the Cities that start with a B?
*/

SELECT
    BillingCity,
    round(avg(total),2)
FROM
    Invoice
WHERE -- ALWAYS before the Group By Clause
    BillingCity like 'B%'
GROUP BY
    BillingCity
HAVING -- ALWAYS after the Group BY Clause
    avg(total)> 5.00
ORDER By
    BillingCity
```

BillingCity	round(avg(total),2)
Bangalore	6.11
Berlin	5.37
Bordeaux	5.66

Grouping by many fields

```
Description: Grouping in SQL |What are the average invoice totals by billing country and by city?
*/
SELECT
    BillingCountry,
    BillingCity,
    round(avg(total),2)
FROM
    Invoice
WHERE
    BillingCountry = 'Canada'
GROUP BY
    BillingCountry,
    BillingCity
ORDER By
    BillingCountry
```

BillingCountry	BillingCity	round(avg(total),2)
Canada	Edmonton	5.37
Canada	Halifax	5.37
Canada	Montréal	5.66